

NeoGuard
AI-Powered Neonatal ICU
Monitoring System
Project Report

Submitted by

ARUN KUMAR P

Bachelor of Engineering

Electronics and Communication Engineering

1. Executive Summary

NeoGuard is an intelligent neonatal monitoring system designed to enhance patient safety in Neonatal Intensive Care Units (NICUs) by reducing false alarms and improving early detection of critical medical conditions. Conventional bedside monitors often generate a large number of alerts, many of which are false positives caused by sensor noise, patient movement, or temporary physiological fluctuations. Studies indicate that up to 87% of NICU alarms are false, leading to alarm fatigue among healthcare staff and delayed responses to genuine emergencies.

NeoGuard addresses this challenge by integrating Internet of Things (IoT) sensors, real-time data processing, and artificial intelligence–based anomaly detection. The system continuously monitors key infant vital parameters such as heart rate (HR), respiratory rate (RR), oxygen saturation (SpO₂), temperature, and blood pressure. A five-stage AI decision pipeline evaluates incoming sensor data to filter noise, predict anomalies, apply clinical decision rules, and generate alerts only when medically meaningful conditions are detected.

By combining low-cost hardware with advanced AI techniques, NeoGuard provides a scalable, cost-effective, and reliable monitoring solution for NICUs. The system also includes a real-time web dashboard and mobile alert system to ensure healthcare professionals receive timely and actionable notifications.

2. Abstract

Neonatal Intensive Care Units (NICUs) require continuous monitoring of premature infants whose physiological parameters can change rapidly and unpredictably. Conventional bedside monitoring systems rely on simple threshold-based alarms, which often generate a large number of alerts. Studies have shown that a significant portion of these alarms—sometimes up to 80–87%—are false positives caused by sensor noise, patient movement, or temporary fluctuations in vital signs. This excessive number of alerts can lead to alarm fatigue, where healthcare professionals may become desensitized to alarms, potentially delaying response to genuine medical emergencies.

This project presents NeoGuard — an AI-powered neonatal ICU monitoring system designed to reduce false alarms and improve the reliability of critical alerts. The system integrates IoT-based biomedical sensors, real-time data processing, and an intelligent multi-stage decision pipeline to monitor key infant vital parameters such as heart rate (HR), respiratory rate (RR), oxygen saturation (SpO₂), temperature, and blood pressure. The collected physiological data is processed through a five-stage AI pipeline that includes signal quality filtering, anomaly detection using time-series prediction models, clinical decision rule evaluation, persistence validation, and alert suppression mechanisms.

NeoGuard also incorporates a web-based dashboard and mobile notification system, enabling healthcare professionals to monitor patients in real time and receive timely alerts when abnormal physiological conditions are detected. By combining low-cost hardware with advanced artificial intelligence techniques, the proposed system aims to provide a cost-

effective, scalable, and intelligent neonatal monitoring solution that enhances patient safety and supports faster medical intervention in NICU environments.

3. Introduction

Premature infants admitted to Neonatal Intensive Care Units (NICUs) require continuous monitoring of their vital signs because even minor physiological changes can rapidly develop into life-threatening conditions. Healthcare professionals rely heavily on bedside monitoring systems to track parameters such as heart rate, oxygen saturation, respiratory rate, and temperature.

However, traditional monitoring systems operate using simple threshold-based alarms that trigger alerts whenever a parameter crosses a predefined limit. While this approach helps detect abnormalities, it also produces a large number of unnecessary alerts. Excessive alarms can overwhelm healthcare staff, resulting in alarm fatigue, where clinicians may unintentionally delay responses due to frequent false alarms.

NeoGuard is developed to address this critical issue by implementing AI-driven monitoring and intelligent alert filtering. The system continuously analyzes real-time patient data streams and evaluates them using machine learning models and clinically validated decision rules. Instead of relying on a single threshold, NeoGuard considers multiple vital parameters simultaneously and ensures that abnormal conditions persist for a defined duration before triggering an alert.

This approach significantly reduces false alarms while ensuring that genuine medical emergencies are detected promptly.

4. Scope of the Project

The scope of the NeoGuard — AI-Powered Neonatal ICU Monitoring System focuses on developing an intelligent and reliable monitoring solution for neonatal patients in Intensive Care Units. The system aims to enhance traditional monitoring methods by integrating Internet of Things (IoT) devices, artificial intelligence, and real-time data visualization to provide accurate and timely alerts for critical medical conditions.

The project involves the design and implementation of a system capable of continuously monitoring key neonatal vital parameters such as heart rate (HR), respiratory rate (RR), oxygen saturation (SpO₂), body temperature, and blood pressure. These physiological signals are collected using biomedical sensors connected to an embedded microcontroller and transmitted to a backend server for processing.

The scope also includes the development of an AI-based decision framework that analyzes real-time sensor data using anomaly detection models and clinically validated rules. This approach ensures that alerts are generated only when abnormal physiological patterns persist, thereby reducing the number of false alarms commonly observed in conventional NICU monitoring systems.

Additionally, the project covers the development of a web-based dashboard that allows healthcare professionals to monitor patient vitals in real time, review alert histories, and respond quickly to critical situations. The system also supports mobile notification mechanisms that instantly inform doctors and nurses when a potential medical emergency is detected.

Overall, the project aims to provide a cost-effective, scalable, and intelligent neonatal monitoring platform that can assist healthcare professionals in improving patient safety, reducing alarm fatigue, and enabling early detection of life-threatening conditions in neonatal intensive care environments.

5. Software and Hardware Requirements

Hardware Requirements

The NeoGuard system requires a set of hardware components for collecting neonatal vital signs and transmitting them to the monitoring system. These components ensure accurate sensing, processing, and communication of physiological data.

- **ESP8266 Microcontroller** – Acts as the main embedded device responsible for reading sensor data and transmitting it to the backend server via Wi-Fi.
- **MAX30102 Pulse Oximeter Sensor** – Measures vital parameters such as heart rate (HR) and oxygen saturation (SpO₂).
- **DHT11 Temperature Sensor** – Used to monitor the body temperature of the infant.
- **Power Supply Module** – Provides stable power to the microcontroller and sensors.
- **Connecting Wires and Breadboard/PCB** – Used for circuit connections and hardware integration.
- **Computer or Laptop** – Used for system development, running the backend server, and monitoring data through the dashboard.
- **Wi-Fi Network** – Required for transmitting sensor data from the ESP32 device to the backend system.

Software Requirements

The software components are responsible for data processing, AI analysis, system control, and user interface development.

- **Python (Version 3.11 or higher)** – Used for backend development and data processing.
- **FastAPI Framework** – Provides high-performance backend APIs for handling real-time sensor data and system communication.
- **React.js** – Used for developing the frontend dashboard to visualize patient vitals and alerts.
- **Tailwind CSS** – Used for designing responsive and user-friendly interfaces.

- **WebSocket Protocol** – Enables real-time data streaming between the backend server and the frontend dashboard.
- **Chronos-Bolt AI Model** – Used for time-series anomaly detection and prediction of abnormal vital patterns.
- **Groq LLaMA 3.3 Model** – Used to generate explainable alert messages for healthcare professionals.
- **Telegram Bot API** – Used to send instant notifications and alerts to doctors and nurses.
- **Docker (Optional)** – Used for containerized deployment of the system.
- **Database (Future Implementation: PostgreSQL with TimescaleDB)** – Used for storing historical patient data and alert records.

6. System Design and Implementation

The **NeoGuard — AI-Powered Neonatal ICU Monitoring System** is designed as a modular and scalable architecture that integrates **IoT sensors, artificial intelligence, and real-time monitoring technologies**. The system is structured into multiple layers that work together to ensure accurate monitoring of neonatal vital parameters and reliable detection of abnormal physiological conditions.

1. System Design

The NeoGuard system follows a **layered architecture** consisting of three major components:

1. Hardware Data Acquisition Layer
2. Backend Processing and AI Analysis Layer
3. Frontend Monitoring and Alert Layer

This architecture enables efficient collection, processing, and visualization of neonatal health data in real time.

Hardware Data Acquisition Layer

The hardware layer is responsible for collecting physiological data from neonatal patients. Biomedical sensors are connected to an **ESP32 microcontroller**, which acts as the primary embedded device.

The following sensors are used in the system:

- **MAX30102 Pulse Oximeter Sensor** – Measures heart rate (HR) and oxygen saturation (SpO₂).
- **DTH11 Temperature Sensor** – Monitors the body temperature of the infant.

The ESP32 continuously reads sensor data and transmits the readings to the backend server through a **Wi-Fi network** at regular intervals (approximately one reading per second). This enables continuous monitoring of neonatal vital parameters.

Backend Processing and AI Analysis Layer

The backend system is developed using **Python and FastAPI**, which provides a high-performance framework for handling real-time data streams and API communication.

All incoming sensor readings pass through a **five-stage intelligent processing pipeline** designed to reduce false alarms and detect genuine medical emergencies.

Gate 1 – Signal Quality Index (SQI)

This stage evaluates the reliability of sensor readings and removes data affected by noise, motion artifacts, or signal interruptions. Only high-quality signals are forwarded to the next stage for analysis.

Gate 2 – AI-Based Anomaly Detection

NeoGuard utilizes the **Chronos-Bolt time-series prediction model** to analyze patterns in recent sensor readings. The model predicts upcoming values and compares them with actual readings. Significant deviations indicate potential anomalies.

Gate 3 – Clinical Decision Rule Evaluation

The system evaluates multiple vital parameters simultaneously using **clinically validated decision rules**. These rules consider combinations of vital signs such as heart rate, respiratory rate, oxygen saturation, and temperature. The thresholds are adjusted according to the infant's **corrected gestational age**, ensuring medically appropriate alert conditions.

Gate 4 – Persistence Filter

To prevent alerts caused by temporary fluctuations, abnormal conditions must persist for a predefined duration before the system confirms an alert.

Gate 5 – Cooldown Suppression

Once an alert is triggered, the system temporarily suppresses repeated alerts for the same condition to avoid excessive notifications and reduce alarm fatigue.

Frontend Monitoring and Alert Layer

The frontend interface is developed using **React, TailwindCSS, and WebSocket communication** to provide real-time monitoring of neonatal patients.

The dashboard displays:

- Real-time vital sign graphs
- Active alert notifications
- Patient monitoring panels
- Historical alert logs
- Patient health reports

The WebSocket protocol ensures that sensor data is streamed to the dashboard instantly, allowing healthcare professionals to monitor patient conditions continuously.

2. Alert and Notification System

When the system detects a critical condition, NeoGuard automatically generates alerts and sends notifications to healthcare professionals.

The system integrates with:

- **Telegram Bot API**
- **WhatsApp notifications using Twilio**

Alerts include a **plain-language explanation of the detected medical condition**, enabling doctors and nurses to quickly understand the situation and take appropriate action.

3. Data Management

Currently, NeoGuard processes real-time patient data using **in-memory storage** for rapid access and processing. In future implementations, the system will integrate **PostgreSQL with TimescaleDB** for long-term storage and analysis of neonatal health data.

This will allow hospitals to maintain historical records of patient vitals and perform advanced data analytics.

4. System Performance

The NeoGuard system is designed to operate efficiently with minimal hardware requirements. The backend service consumes approximately **50 MB of memory**, while the AI model requires around **200 MB of memory** in its standard configuration. The system can support multiple neonatal patients simultaneously while maintaining real-time performance.

Results and Discussion

The NeoGuard system was successfully designed and implemented as a prototype for intelligent neonatal monitoring in NICU environments. The system integrates IoT-based

biomedical sensors, artificial intelligence models, and a web-based monitoring dashboard to continuously track neonatal vital parameters and generate meaningful alerts.

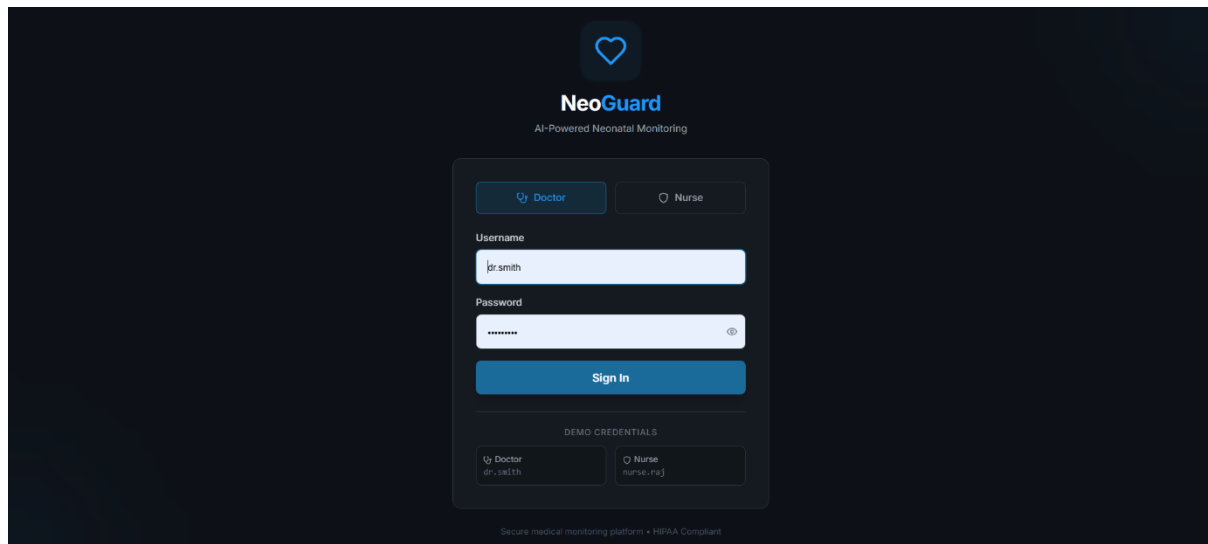
During testing, simulated patient data was used to evaluate the functionality of the system. The sensors successfully captured physiological parameters such as **heart rate, oxygen saturation, respiratory rate, and body temperature**, which were transmitted to the backend server using the ESP32 microcontroller. The backend processing system handled the incoming data efficiently and processed it through the **five-stage AI pipeline**, including signal quality filtering, anomaly detection, clinical decision rule evaluation, persistence filtering, and cooldown suppression.

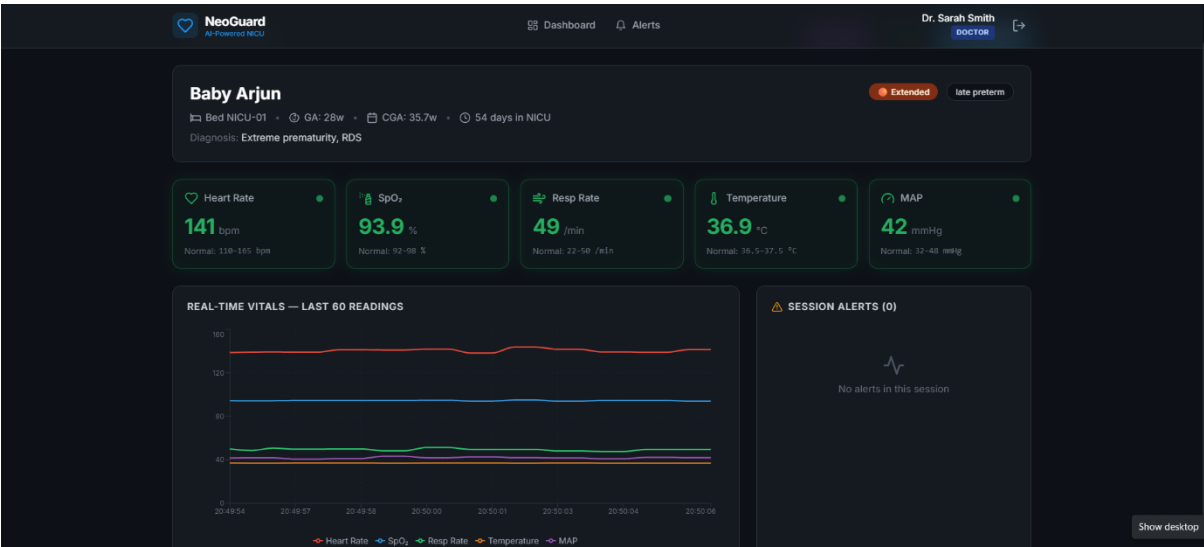
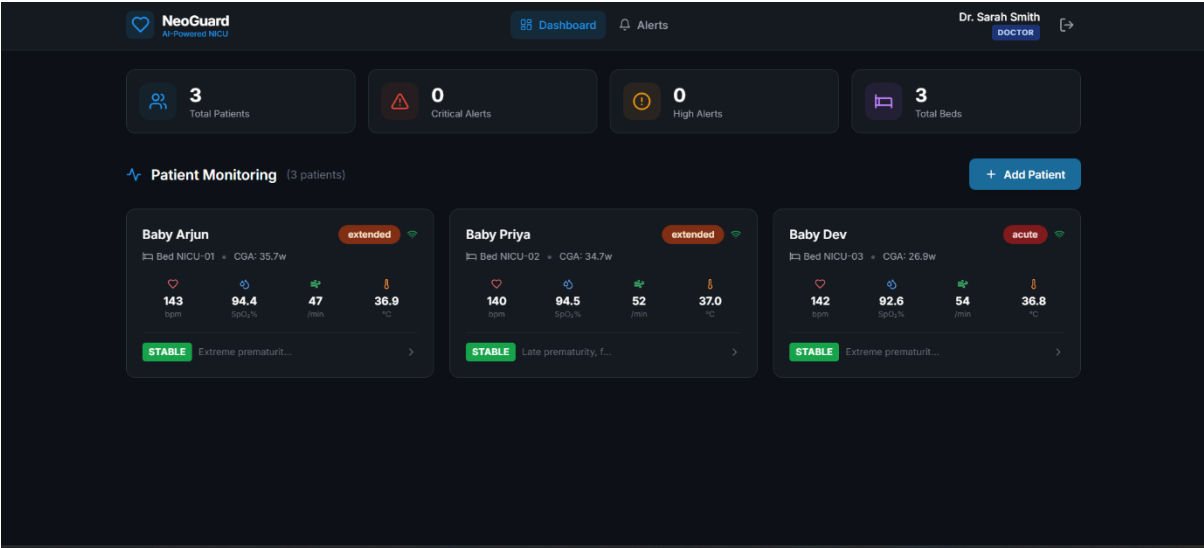
The system demonstrated the ability to **identify abnormal physiological patterns** such as apnea-bradycardia-desaturation events, hypoxemia, and early sepsis indicators. By combining multiple vital parameters and applying persistence checks, NeoGuard significantly reduced the number of unnecessary alerts compared to traditional threshold-based monitoring systems.

The **React-based dashboard** successfully displayed real-time patient vitals using WebSocket communication, allowing healthcare professionals to monitor patient status continuously. Alert notifications were delivered through mobile communication platforms such as Telegram and WhatsApp, ensuring immediate awareness of critical events.

Appendices:

Doctor Dashboard:





Nurse Dashboard:

